THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2^8	Add the exponents: $5+3=8$
2	**(a)** 10^{10}	Add the exponents: $6+4=10$
3	**(a)** 7^7	Subtract the exponents
4	**(d)** 3^8	Multiply the exponents
5	**(b)** $\frac{1}{25}$	$a^{-n}=\frac{1}{a^n}$
6	**(c)** 1	Any non-zero number to the power 0 is 1
7	**(d)** 7.2×10^4	One non-zero digit before the decimal point
8	**(a)** 1.5×10^8	One non-zero digit before the decimal point
9	**(c)** 10^9	A thousand million
10	**(d)** 9	Reciprocal squared

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Add exponents: $2^{5-3}=2^2=4$ (2).	2
12	$\frac{1}{5^3}=\frac{1}{125}$ (2).	2
13	4.5×10^{-4} (2).	2
14	620000 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Add in the numerator: 2^7 (1); subtract: 2^2 (1); value 4 (1).	1+1+1
16	Add in the numerator: 3^7 (1); subtract: 3^1 (1); value 3 (1).	1+1+1
17	Add in the numerator: 5^7 (1); subtract: 5^5 (1); value 3125 (1).	1+1+1
	OR: Add in the numerator: 7^6 (1); subtract: 7^3 (1); value 343 (1).	1+1+1
18	Add in the numerator: 10^{13} (1); subtract: 10^9 (1); value 1000000000 (1).	1+1+1
	OR: Add in the numerator: 2^{11} (1); subtract: 2^9 (1); value 512 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) a^{m+n}, a^{m-n}, a^{mn} (3). (ii) $=2^7\times5^4/(2^3\times5^3)=2^4\times5=80$ (2).	3+2
	OR: (i) 0.00047 (1). (ii) 9.2×10^{-7} (2). (iii) $35000=3.5\times10^4$ (2).	1+2+2
20	(i) $\left(\frac{2}{3}\right)^{-1}=\frac{3}{2}$ (2). (ii) $5^6\times5^2/5^{10}=5^{-2}=\frac{1}{25}$ (3).	2+3
	OR: (i) $2\times10^9\times3^3=5.4\times10^{10}$ (2). (ii) 5.4×10^{10} (1). (iii) after 4 hours: 1.62×10^{11} (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 2: **Power Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3^{10}	Add the exponents: $4 + 6 = 10$
2	**(c)** 2^5	Subtract the exponents
3	**(b)** 1	Subtract the exponents
4	**(d)** 10^{15}	Multiply the exponents
5	**(c)** $\frac{1}{1000}$	$a^{-n} = \frac{1}{a^n}$
6	**(a)** $2^4 \times 3^4$	$(ab)^n = a^n b^n$
7	**(b)** 3.5×10^{-3}	One non-zero digit before the decimal point
8	**(d)** 4.12×10^{-2}	One non-zero digit before the decimal point
9	**(d)** 1000	$2^{10} = 1024$
10	**(a)** $\frac{1}{25}$	$5^{6-8} = 5^{-2}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$3^8 \div 3^6 = 3^2 = 9$ (2).	2
12	-1 for an odd power (1); 1 for an even power (1).	1+1
13	$1 + \frac{1}{4} = \frac{5}{4}$ (2).	2
14	Add exponents: $a^m \times a^n = a^{m+n}$ (1); e.g. $5^2 \times 5^3 = 5^5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Add in the numerator: 3^{10} (1); subtract: 3^6 (1); value 729 (1).	1+1+1
16	Add in the numerator: 5^8 (1); subtract: 5^5 (1); value 3125 (1).	1+1+1
17	Add in the numerator: 4^8 (1); subtract: 4^6 (1); value 4096 (1).	1+1+1
	OR: Add in the numerator: 6^8 (1); subtract: 6^3 (1); value 216 (1).	1+1+1
18	Add in the numerator: 2^{11} (1); subtract: 2^7 (1); value 128 (1).	1+1+1
	OR: Add in the numerator: 3^{14} (1); subtract: 3^{12} (1); value 531441 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2^0 \div 2^3 = 2^{-3} = \frac{1}{8}$ (2). (ii) $\frac{9}{8} \times 6 = \frac{27}{4}$ (3).	2+3
	OR: (i) $9.46 \times 10^{12} \times 10^5 = 9.46 \times 10^{17}$ km (3). (ii) Already in standard form (2).	3+2
20	(i) $(1 + \frac{1}{4}) \times 4 = 5$ (2). (ii) $x - 4 = 5$, $x = 9$ (3).	2+3
	OR: (i) 1.4×10^9 and 3.29×10^6 (2). (ii) ≈ 426 per km² (2). (iii) City is about 26 times denser (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.